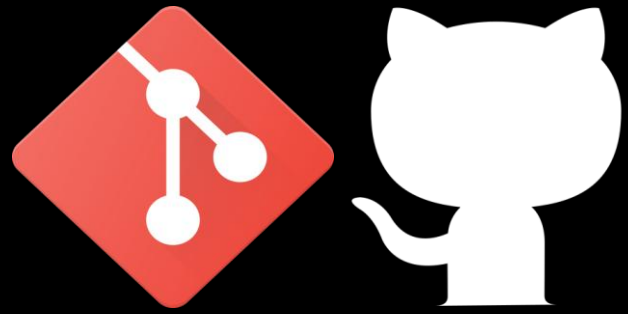
GIT & GITHUB



DEVLABS 2.0

WHAT WE WILL DISCUSS ?

- What is Git ?
- What is GitHub?
- Git vs GitHub
- Git Workflow
- Setting up Github Account
- Setting up Git Account
- Git Commands
- Github Repository
- Git and Github Task for Session

GIT

Version Control System is a tool that helps to track changes in code

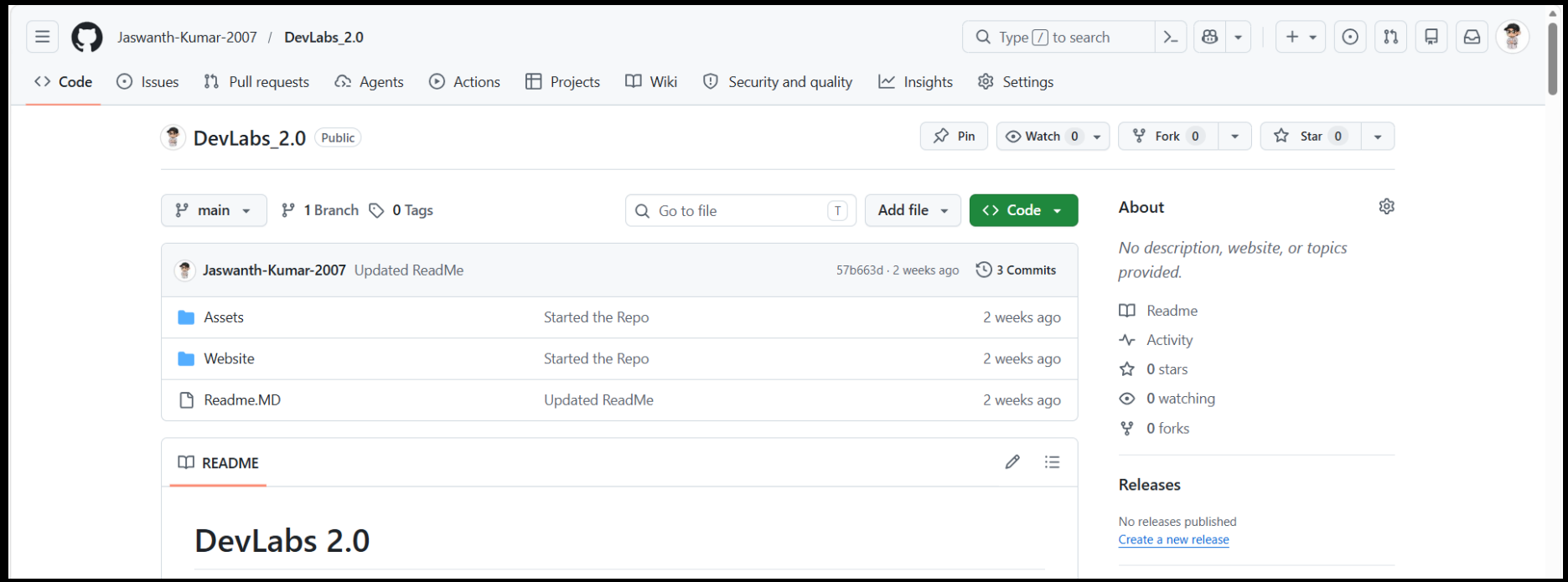
Git is a Version Control System . It is :

- Popular
- Free & Open Source
- Fast and Scalable
- Saving project history
- Tracking changes
- Collaboration

GITHUB

Website that Allows Developers to store and Manage their code using Git

https://github.com



Why Developers Use GitHub ?

1. Collaboration

Multiple developers can work on the same project together.

- Share code
- Review code
- Work in teams

2. Backup and Cloud Storage

Projects are stored online safely.

Benefits:

- Prevents data loss
- Access project from anywhere

3. Version Control

GitHub stores project history.

Developers can:

- Track changes
- Restore old versions
- Compare updates

4. Open Source Contributions

Developers contribute to public projects.

Examples:

- Fix bugs
- Add features
- Learn from real projects

5. Portfolio for Developers

GitHub acts like a coding portfolio.

Developers showcase:

- Projects
- Skills
- Contributions

6. Team Management

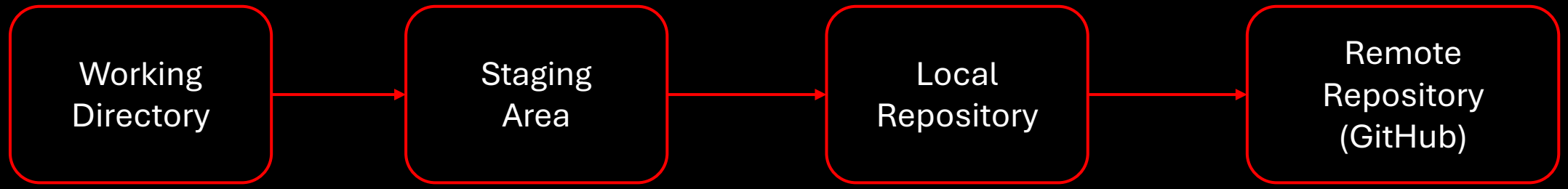
GitHub helps teams manage projects using:

- Pull Requests
- Issues
- Branches
- Code Reviews

GIT Vs GITHUB

Git	GitHub
Works in a decentralized architecture where every developer has a full copy of the repository.	Works on a centralized hosting model for sharing and managing repositories online.
Operates mainly via command-line interface (CLI) with optional GUI tools.	Provides a web-based UI + integrations along with Git command support.
Installed and runs on the local machine.	Hosted on the cloud (web platform).
Focuses on version control operations like commit, branch, merge and rebase.	Focuses on collaboration workflows like pull requests, issues and code reviews.
Maintains complete history and snapshots of the project locally.	Hosts repositories and manages team collaboration and access control.
Open-source and maintained by the community (originally by Linus Torvalds).	Owned and maintained by Microsoft.
Does not include built-in access control or user management.	Provides role-based access control and permissions.
Competes with SVN, Mercurial, CVS.	Competes with GitLab, Bitbucket, Azure DevOps.

GIT Workflow

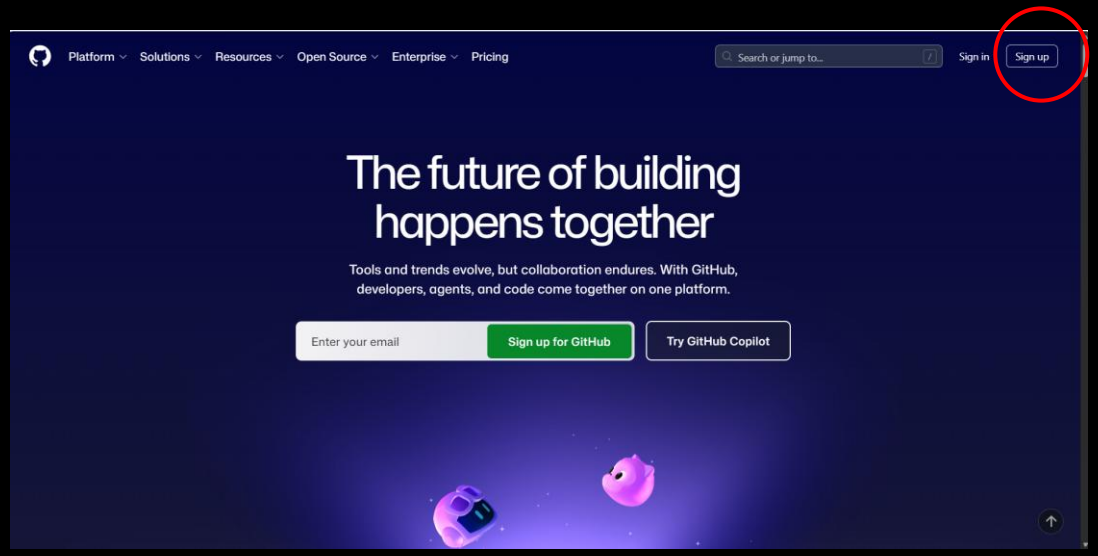


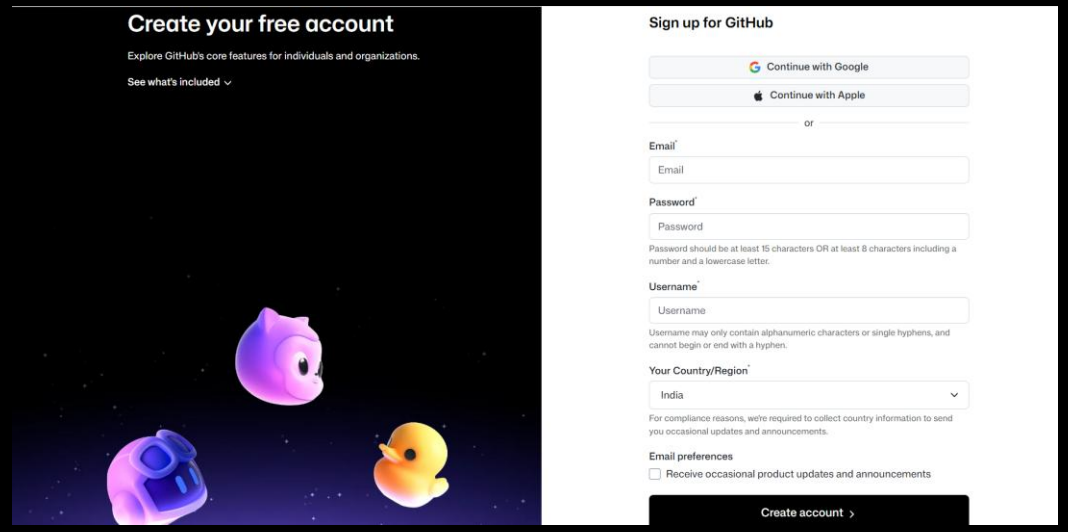
- Working Directory - Where you make changes
- git add - Stage changes
- git commit - Save changes to your repository
- git push - Share changes with others
- git status - Check what's going on
- Undo/Amend - Fix mistakes (`git restore`, `git reset`, `git commit --amend`)

Workflow Diagram

```
[Working Directory] --git add--> [Staging Area] --git commit--> [Repository]
```

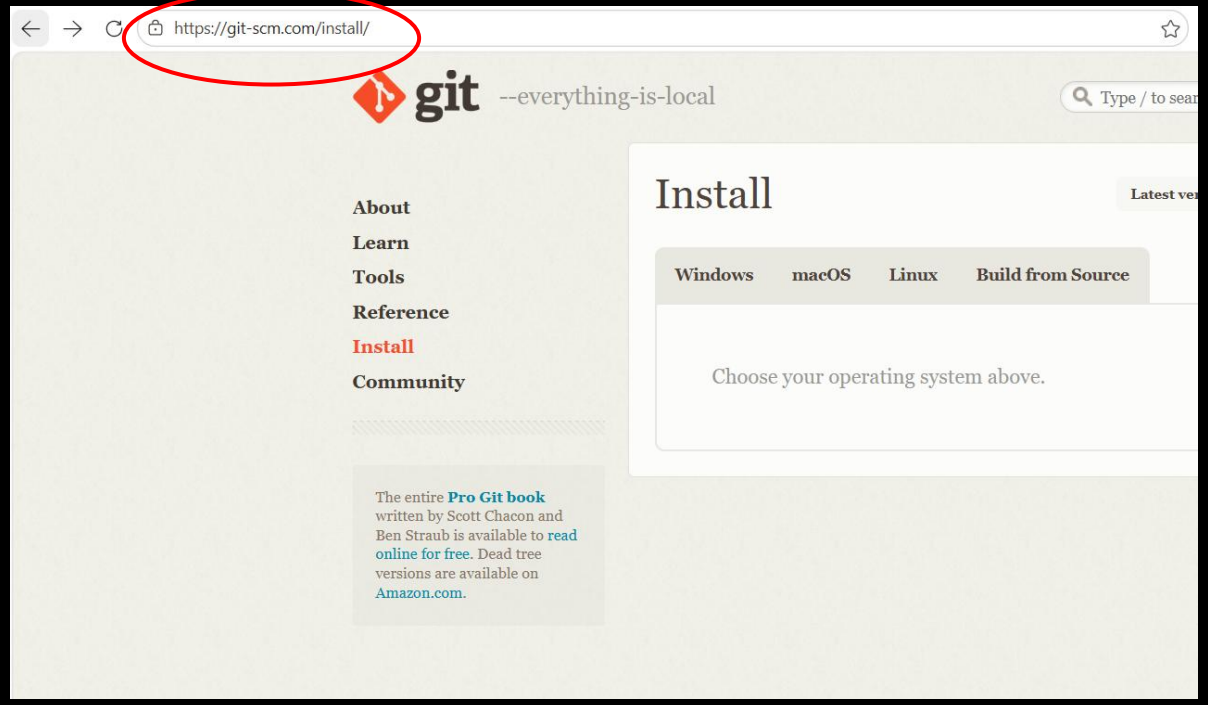
SETTING UP GITHUB ACCOUNT





SETTING UP GIT

https://git-scm.com/install/



SETTING UP GIT

Check Git is Successfully Installed on your Device

git --version

Visual Studio Code

Windows (Git Bash)

Mac (Terminal)

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kamir>git --version
git version 2.52.0.windows.1
```

CONFIGURING GIT

Next , Open VS Code Terminal

git config –global user.name “Your Name”

git config –global user.email youremail@gmail.com

git config --list

Git uses your name and email to label your commits.

GIT COMMANDS

git init : Initialize a new Git Repository in the current directory

```
git init
```

What Happens When You Run git init ?

- Git creates a hidden folder called .git inside your project.
- This is where Git stores all the information it needs to track your files and history.

git status : Display the state of the working directory & staging area

```
git status
```

GIT COMMANDS

What is an Untracked file

- An **untracked file** is any file in your project folder that Git is not yet tracking.
- These are files you've created or copied into the folder, but haven't told Git to watch.

What is a Tracked file

- A **tracked file** is a file that Git is watching for changes.
- To make a file tracked, you need to add it to the staging area

GIT COMMANDS

Staging Environment

- `git add <file>` - Stage a file
- `git add --all` or `git add -A` - Stage all changes
- `git status` - See what is staged
- `git restore --staged <file>` - Unstage a file

git add : Add Changes to the staging area

git add file.txt

How to unstage a File

git restore –staged index.html

git reset HEAD index.html

GIT COMMANDS

git commit : Record Changes to the Repository

```
git commit –m “Initial Commit”
```

- `git commit -m "message"` - Commit staged changes with a message
- `git commit -a -m "message"` - Commit all tracked changes (skip staging)
- `git log` - See commit history

Commit all changes with staging (-a)

```
git commit -a -m "message"
```

Note: git commit -a does not work for new/untracked files. You must use git add <file> first for new files.

GIT COMMANDS

Other Useful Commit Options

- **Create an empty commit:**
  `git commit --allow-empty -m "Start project"`
- **Use previous commit message (no editor):**
  `git commit --no-edit`
- **Quickly add staged changes to last commit, keep message:**
  `git commit --amend --no-edit`

git log : Shows the Commit History

- `git log` - Show full commit history
- `git log --oneline` - Show a summary of commits
- `git show <commit>` - Show details of a specific commit
- `git diff` - See unstaged changes
- `git diff --staged` - See staged changes

GIT COMMANDS

git diff : different between your working directory and the last commit (unstaged changes)

git diff --staged : different between your staged files and the last commit

git diff <commit1> <commit2> : git diff 1234567 89abcde

git log –author=“Alice” :

```
git log --author="Alice"
commit 1a2b3c4d5e6f7g8h9i0j
Author: Alice
Date:   Mon Mar 22 10:12:34 2021 +0100

    Add about page
```

GIT COMMANDS

Undo a Commit

Step 1: Initial Code

`app.js`

JavaScript

```javascript
console.log("Hello");
```

Commit it:

Bash

```bash
git add .
git commit -m "Initial commit"
```

Step 2: Add Dark Mode

Now change code:

JavaScript

```javascript
console.log("Hello");

console.log("Dark mode added");
```

Commit again:

Bash

```bash
git add .
git commit -m "Added dark mode"
```

GIT COMMANDS

Undo a Commit

Current History

```
Commit 1 → Initial commit
Commit 2 → Added dark mode
```

Current file:

JavaScript

```javascript
console.log("Hello");

console.log("Dark mode added");
```

Step 3: Revert Commit 2

Check log:

Bash

```
git log
```

Suppose commit id is:

```
abc123 Added dark mode
```

Now run:

Bash

```
git revert abc123
```

GIT COMMANDS

Undo a Commit

Git creates NEW commit:

```
Commit 1 → Initial commit
Commit 2 → Added dark mode
Commit 3 → Revert "Added dark mode"
```

Final Code After Revert

JavaScript

```
console.log("Hello");
```

Can Use Git Reset

`git reset --hard`

After reset:

```
Commit 1 → Initial commit
```

Code:

JavaScript

```
console.log("Hello");
```

Commit 2 completely disappears.

GIT COMMANDS

git init : Initialize a new Git Repository in the current directory

```
git init
```

What Happens When You Run git init ?

- Git creates a hidden folder called .git inside your project.
- This is where Git stores all the information it needs to track your files and history.

git status : Display the state of the working directory & staging area

```
git status
```

GIT COMMANDS

git tagging

A **tag** in Git is like a label or bookmark for a specific commit.

- `git tag <tagname>` - Create a lightweight tag
- `git tag -a <tagname> -m "message"` - Create an annotated tag
- `git tag <tagname> <commit-hash>` - Tag a specific commit
- `git tag` - List tags
- `git show <tagname>` - Show tag details

Push Tags to Remote

```
git push origin v1.0
```

Delete Tags

```
git tag -d v1.0
```

Locally

```
git push origin --delete tag v1.0
```

Remote Repository

GIT COMMANDS

git stashing

git stash lets you save your uncommitted changes and return to a clean working directory.

- `git stash` - Stash your changes
- `git stash push -m "message"` - Stash with a message
- `git stash list` - List all stashes
- `git stash branch <branchname>` - Create a branch from a stash

git stash apply	Restore your most recent stashed changes (keeps the stash in the stack)
git stash pop	Apply the latest stash **and remove it from the stack**
git stash drop	Delete a specific stash when you no longer need it
git stash clear	Delete all your stashes at once

```
git stash drop stash@{0}
Dropped stash@{0} (abc1234d5678)
```

GIT COMMANDS

git branching

In Git, a branch is like a separate workspace where you can make changes and try new ideas without affecting the main project. Think of it as a "parallel universe" for your code.

git branch feature-branch	Lists or create a new branch
git checkout feature-branch	Switches between branches or Restore files
git switch feature-branch	Switch Branches
git merge feature-branch	Combine changes from one branch into another
git rebase main	move your current branch on top of another branch (e.g., update your feature branch with latest main)
git branch –d feature-branch	Delete Branch

GIT COMMANDS

GIT BRANCHING

- git branch (to check branch)
- git branch -M main (to rename branch)
- git checkout <-branch name-> (to navigate)
- git checkout -b <-new branch-name-> (to create new branch)
- git branch -d <-branch name-> (to delete branch)

GIT COMMANDS

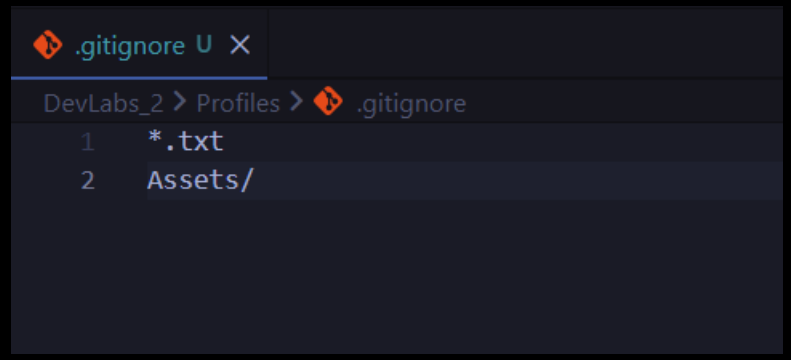
Remote Repositories

`git clone https://github.com/user/repo.git`	Clones a remote repository to the local machine
`git pull origin main`	Fetches and merges changes from a remote repository
`git fetch origin`	Download changes from a repository without merging
`git remote -v`	List the url of the remote repositories

GIT COMMANDS

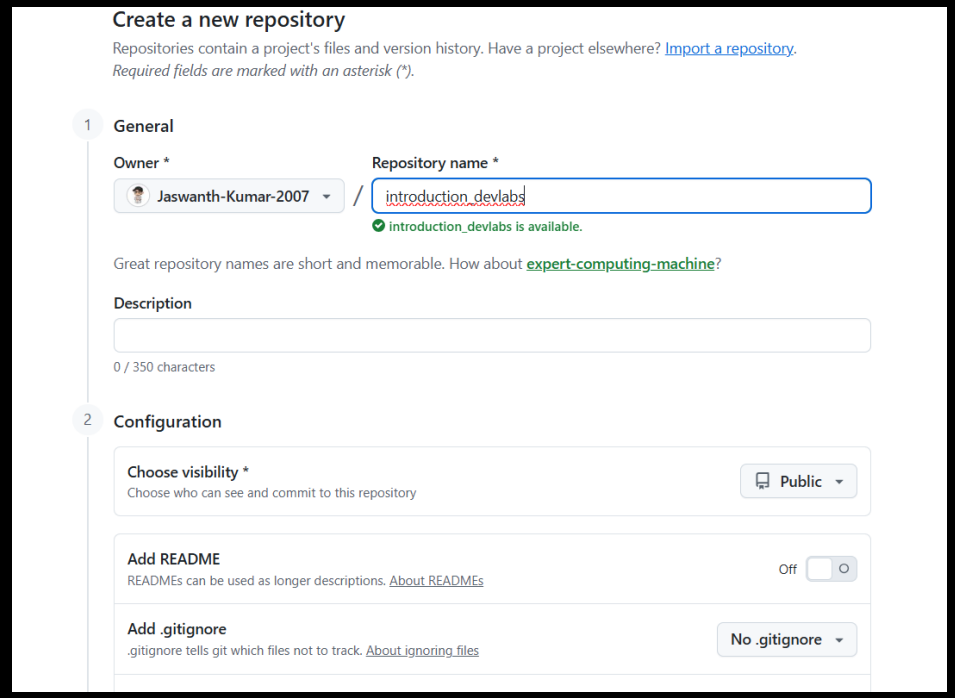
GIT IGNORE FILE

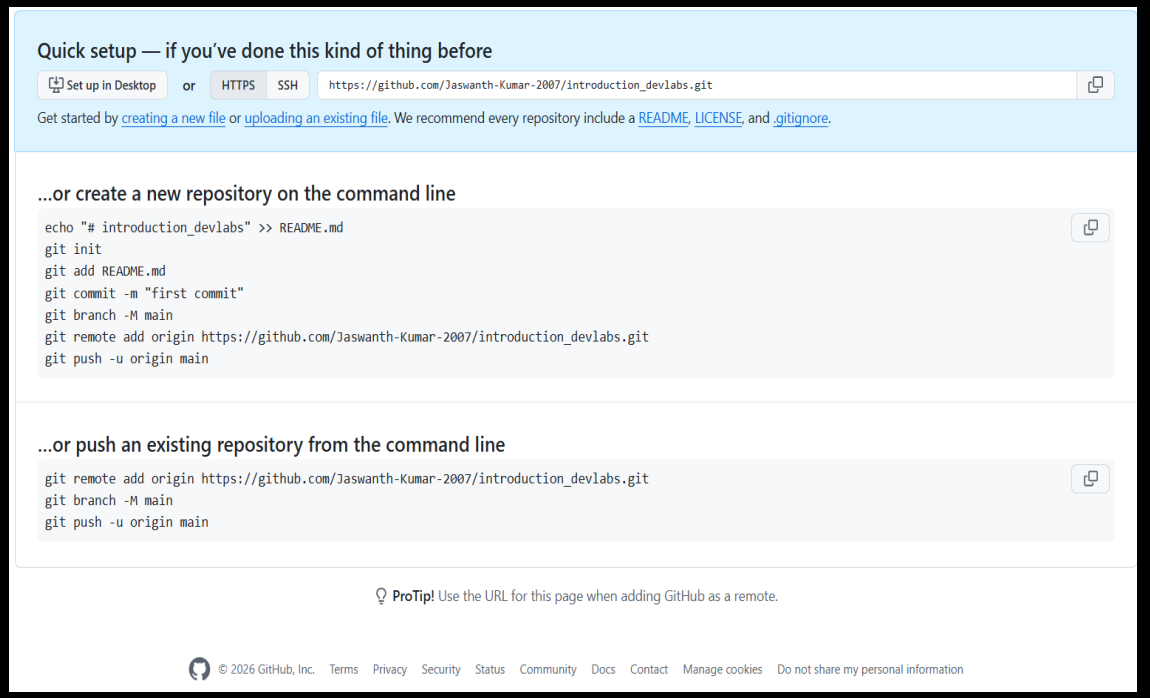
- Create .gitignore file
- Ignore unnecessary files.




```
node_modules
.env
*.log
```

GITHUB NEW REPOSITORY

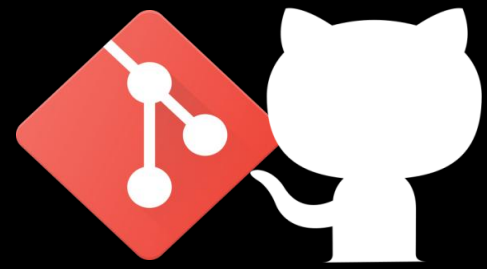




GITHUB NEW REPOSITORY

- git init
- git add README.md
- git commit -m "first commit"
- git branch -M main
- git remote add origin https://github.com/Jaswanth-Kumar2007/introduction_devlabs.git
- git push -u origin main

GIT AND GITHUB TASK for Session



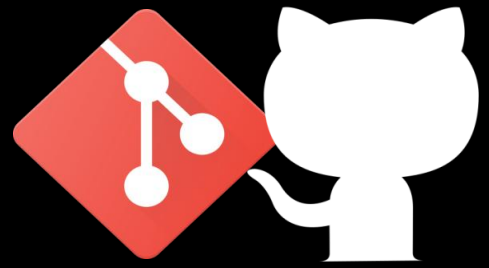
- git clone https://github.com/Jaswanth-Kumar-2007/DevLabs_2.0.git

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Projects\JK Projects\OpenLake\Devlabs>git clone https://github.com/Jaswanth-Kumar-2007/DevLabs_2.0.git
```

- git branch “ID.NO.”

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Projects\JK Projects\OpenLake\Devlabs>git branch B25DS015
```

GIT AND GITHUB TASK for Session



- git checkout “ID.NO.”

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Projects\JK Projects\OpenLake\Devlabs>git checkout B25DS015
```

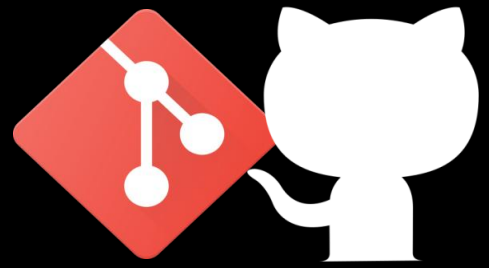
- Create a txt File in the Profile Folder

B25DS015.txt U

DevLabs_2 > Profiles > B25DS015.txt

```
1  Name : Jaswanth Kumar
2  ID No : B25DS015
3  Branch : DSAI
4  Skills : HTML ,CSS , JavaScript , React
```

GIT AND GITHUB TASK for Session



- git add .

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Projects\JK Projects\OpenLake\Devlabs>git add .
```

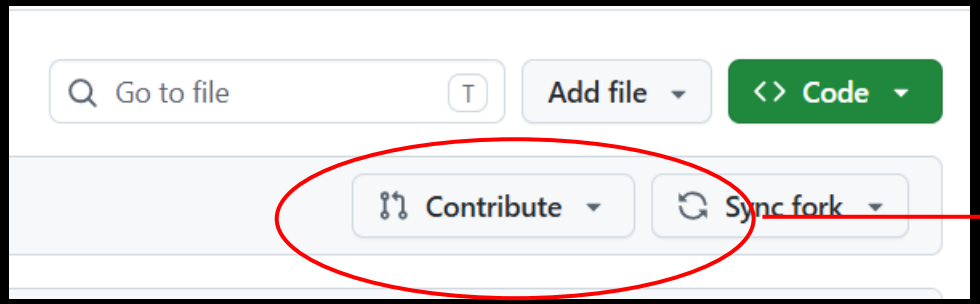
- git commit -m {What Change you Done}

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Projects\JK Projects\OpenLake\Devlabs>git commit -m "Added Profile Txt"
```

GIT AND GITHUB TASK for Session

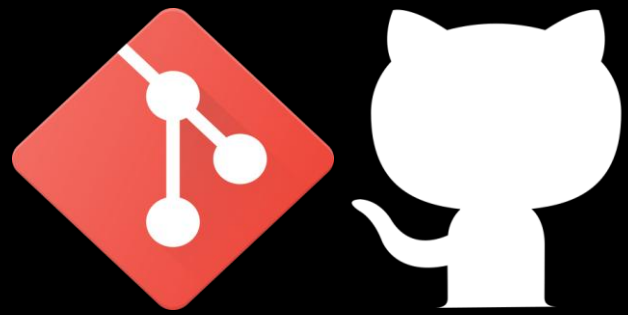
- git push –u origin “ID.NO.”

```
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Projects\JK Projects\OpenLake\Devlabs>git push -u origin B25DS015
```



Push Changes to the Repo

THANK YOU



Again meet in next Session ..... 😊